

STATSIG lecture 1

State space representation of LTI-systems

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

SSC, State Space representation of continuous time SISO LTI-systems

System differential equation for general input $u(t)$ and output $y(t)$

$$\frac{d^n y(t)}{dt^n} + a_{n-1} \frac{d^{n-1} y(t)}{dt^{n-1}} + \dots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_m \frac{d^m u(t)}{dt^m} + \dots + b_0 u(t)$$

External model (unique), transfer function

$$H(s) = \frac{Y(s)}{U(s)} = \frac{b_m s^m + \dots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}$$

Internal model (not unique), SSC

Idea: one n 'th order differential equation $\rightarrow$ n coupled 1'st order differential equations

Choose state vector $x(t) = [x_1(t), x_2(t), \dots, x_n(t)]^T$ ($n \times 1$) vector

Dynamic (SSC) model

$$\begin{aligned} \frac{dx(t)}{dt} &= A x(t) + B u(t) & \text{where } A &= (n \times n) \text{ state transition matrix, } B = (n \times 1) \text{ input matrix} \\ y(t) &= C x(t) + D u(t) & C &= (1 \times n) \text{ output matrix, } D = (1 \times 1) \text{ feedthrough matrix} \end{aligned}$$

Can easily be generalized to MIMO systems, time-varying systems and even non-linear systems (unlike $H(s)$)

SSD, State Space representation of discrete time SISO LTI-systems

System difference equation for general input $u(n)$ and output $y(n)$

$$y(n) = \sum_{i=0}^M b_i u(n-i) - \sum_{i=1}^N a_i y(n-i)$$

External model (unique), transfer function

$$H(z) = \frac{Y(z)}{U(z)} = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + \dots + a_N z^{-N}}$$

Internal model (not unique), SSD

Idea: one N 'th order difference equation $\rightarrow$ N coupled 1'st order difference equations

Choose state vector $x(n) = [x_1(n), x_2(n), \dots, x_N(n)]^T$ ($N \times 1$) vector

Dynamic (SSD) model

$$\begin{aligned} x(n+1) &= A x(n) + B u(n) \\ y(n) &= C x(n) + D u(n) \end{aligned} \quad \text{where} \quad \begin{aligned} A &= (N \times N) \text{ state transition matrix, } B = (N \times 1) \text{ input matrix} \\ C &= (1 \times N) \text{ output matrix, } D = (1 \times 1) \text{ feedthrough matrix} \end{aligned}$$

Can easily be generalized to MIMO systems, time-varying systems and even non-linear systems (unlike $H(z)$)

Equivalence between H(z) and SSD

$$H(z) = C(zI - A)^{-1}B + D \quad \text{and}$$

poles for H(z) = eigenvalues for A

Time-varying systems

$$A \rightarrow A(n), \quad B \rightarrow B(n), \quad C \rightarrow C(n), \quad D \rightarrow D(n)$$

Non-linear systems

$$x(n+1) = f(x(n), u(n))$$
$$y(n) = g(x(n), u(n))$$

MIMO-systems, p inputs & q outputs

$$\begin{aligned} x(n+1) &= A x(n) + B u(n) \\ y(n) &= C x(n) + D u(n) \end{aligned} \quad \text{where}$$

A = (N x N) state transition matrix, B = (N x p) input matrix
C = (q x N) output matrix, D = (q x p) feedthrough matrix